Government Fiscal Spending and Crowding-Out of Private Corporate Investment in India, 1980–2023

An Empirical Investigation Using ARDL Bounds-Testing and Dynamic OLS/ADL Framework

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Abstract

This paper examines the relationship between government fiscal activity and private corporate investment in India over the period 1980–2023, with a focus on whether fiscal expansion crowds out or crowds in private capital formation. Using an ARDL bounds-testing framework, we estimate both long-run and short-run effects across aggregate, sectoral, and industry levels. We find robust evidence that fiscal deficits crowd out private investment in the long run, with a one percent increase in the deficit reducing private investment by roughly 0.4–0.5 percent. In contrast, public investment does not exert a significant independent effect, but becomes complementary to private investment in the post-FRBM reform period. Credit availability emerges as the key short-run driver. These results highlight the importance of fiscal discipline and financial conditions in shaping private investment dynamics in India.

1. Introduction

Private corporate investment is a central driver of long-run economic growth, yet its interaction with government fiscal policy remains theoretically and empirically contested. A long-standing debate in macroeconomics concerns whether government spending stimulates private investment through complementary channels or crowds it out by absorbing scarce financial resources. The answer is likely to depend on institutional context, fiscal conditions, and the composition of public expenditure.

India provides a particularly relevant setting to revisit this question. Over the past four decades, the economy has undergone major structural shifts—including liberalisation in 1991, the enactment of the Fiscal Responsibility and Budget Management (FRBM) Act in 2003, and large-scale public investment initiatives in the post-2019 period. These changes potentially alter the relationship between public and private investment, making it important to account for regime-specific dynamics.

This paper investigates whether government fiscal expansion crowds out or crowds in private corporate investment in India over 1980–2023. We extend the framework of Shankar and Trivedi (2021) by incorporating a longer time horizon, structural breaks, and a reform interaction that allows the investment relationship to vary across policy regimes. Empirically, we use an ARDL bounds-testing approach to examine cointegration and short-run dynamics.

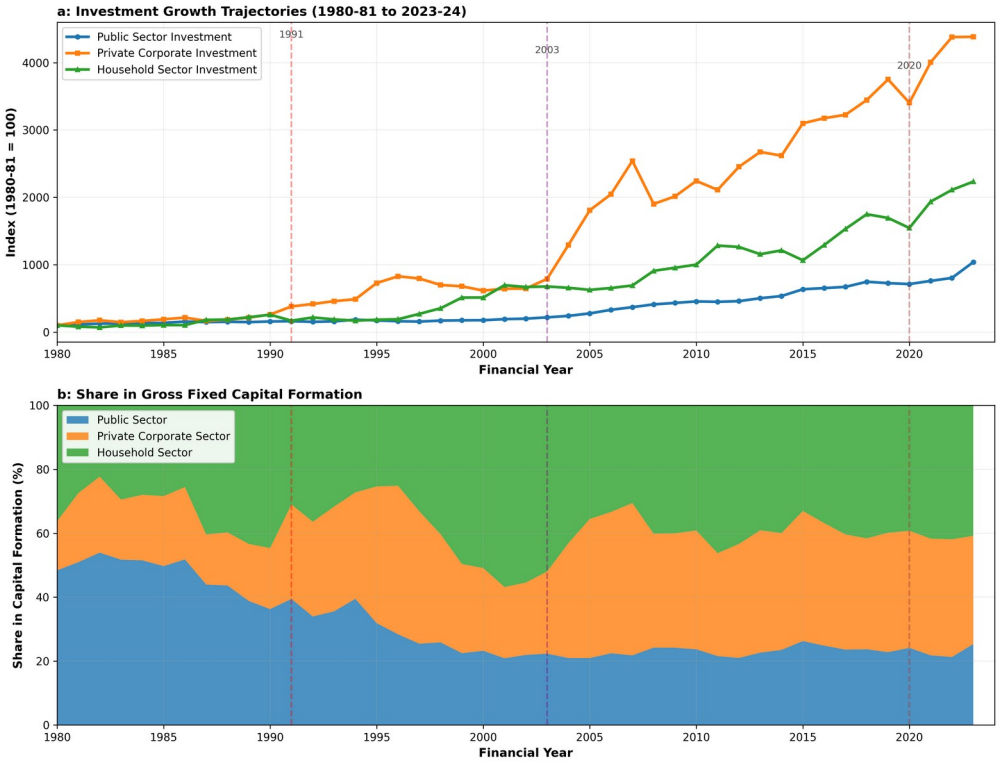
Our main findings are threefold. First, fiscal deficits consistently crowd out private investment in the long run. Second, public investment does not have a significant standalone effect, but becomes complementary in the post-FRBM regime. Third, credit availability is the most important short-run determinant of private investment. Importantly, we find that a stable long-run relationship between public and private investment emerges only after accounting for structural reforms, indicating that the investment nexus in India is inherently regime-dependent.

These findings contribute to the literature by highlighting the role of fiscal governance and financial conditions in shaping investment outcomes. They also carry clear policy implications: sustained fiscal discipline and improved credit access are critical for fostering private capital formation.

2. Investment Dynamics and Stylized Facts

The gross fixed capital formation (GFCF) is the acquisition and creation of assets for their use in the medium to long run, netted against disposals of produced fixed assets. In India's context, understanding the investment dynamics across public and private sectors is crucial, particularly given the dramatic structural transformations the economy has undergone. Over four decades, India's investment landscape reveals distinct phases: the pre-liberalisation period (1980–1991), the post-liberalisation growth era (1991–2003), and the institutional reform and infrastructure expansion period (2003–2023). The decadal correlation coefficients between public and private investment rates reveal initial substitutive effects that gradually transition to complementarity, reflecting the evolving policy environment.

Figure 1. Investment Dynamics



The public sector investment rate exceeded the private sector during the 1980s (48.4 vs 19.5 percent), reflecting the dominance of state-led development in India's relatively closed economy. However, the trend underwent significant reversal following economic liberalisation in 1991, with private sector investment rate rising to 33.7 percent by the 1990s even as public investment declined to 31.9 percent. This transition, coupled with a weak correlation of –0.34, reflects the volatile adjustment period as the economy shifted from a public-sector-led to a private-sector-dominant investment regime. The correlation pattern subsequently improved, turning significantly positive (0.71**) in the 2010s following FRBM-induced fiscal consolidation and the launch of large-scale public infrastructure initiatives.

Table 1. Investment Rate - Stylized Summary

Period$	Mean	Max	Min	Std.Dev	Mean	Max	Min	Std.Dev	Correlation
1980–81 to 1989–90	48.4	53.9	38.8	4.78	19.5	23.8	15.4	3.03	0.69**
1990–91 to 1999–00	31.9	39.5	22.5	6.06	33.7	46.5	19.1	8.07	–0.34
2000–01 to 2009–10	22.3	24.2	20.9	1.25	33.9	47.6	22.3	9.38	–0.01
2010–11 to 2019–20	23.3	26.3	21.0	1.51	36.7	40.7	32.2	2.30	0.71**
2020–21 to 2023–24	23.1	25.3	21.3	1.91	36.0	36.8	33.9	1.38	–0.78
1980–81 to 2023–24	30.7	53.9	20.9	11.02	31.4	47.6	15.4	8.92	–0.68***

*Notes: Public Sector and Private refer to public sector and private corporate sector investment rates respectively. ***, ** and * represent significance at 1%, 5% and 10% level, respectively. $ Financial year of India (April to March). Std.Dev = Standard Deviation.*

The stylized summary statistics in Table 1 reveal several important patterns. First, the volatility of private sector investment (ranging from 3.03 to 9.38 percent standard deviation) substantially exceeds that of public sector investment (1.25 to 6.06 percent), indicating the more cyclical and volatile nature of private corporate capital formation. Second, the correlation between public and private investment varies markedly across decades, shifting from complementary (0.69** in the 1980s) to weakly substitutive or neutral (–0.34 and –0.01 in the 1990s and 2000s) to again complementary (0.71** in the 2010s). This temporal pattern of shifting correlations underscores the critical role of institutional change and macroeconomic regime in shaping the public–private investment nexus. Third, the 1990s represent a structural break, marked by rapid decline in public investment share (from 48.4 to 31.9 percent) and sharp rise in private investment share (from 19.5 to 33.7 percent), reflecting the transition to a market-oriented economy. The 2000s onward show stabilization of the public investment share around 22–23 percent, while private investment stabilizes around 33–37 percent, suggesting a new equilibrium post-FRBM.

Figure 2. Share of Public Sector Investment

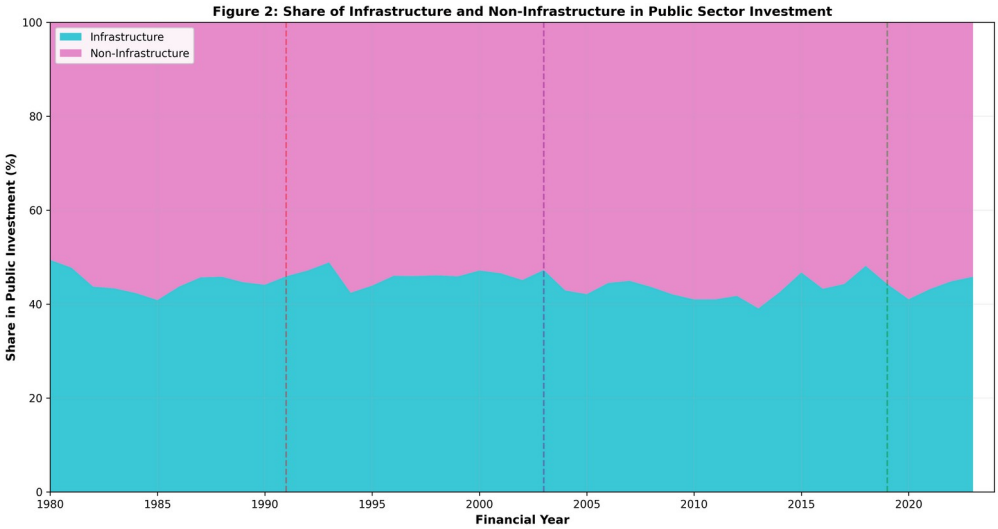


Figure 2 illustrates the widening gap between infrastructure and non-infrastructure public investment, particularly pronounced from the mid-2000s onwards. The growing share of infrastructure investment reflects deliberate policy shifts, most notably the announcement of the National Infrastructure Pipeline (2019) and PM GatiShakti National Master Plan (2021). Infrastructure's share, which ranged from 30–40 percent in earlier decades, has expanded to nearly 45–50 percent in recent years (2019–2023), underscoring the policy prioritisation toward productive public assets. This composition shift is significant for our analysis, as it helps explain potential mechanisms through which fiscal policy may transition from crowding-out to crowding-in, depending on whether government spending targets complementary infrastructure or competes directly with private sector activities.

3. Literature Review

The relationship between public investment and private investment has long been debated within macroeconomic theory, particularly through the lens of the crowding-out versus crowding-in hypothesis. The 'real or direct' crowd-out occurs when an increase in public capital formation displaces private investment by competing for scarce resources, whereas 'financial or indirect' crowd-out arises when government borrowing leads to higher interest rates, thereby discouraging private investment (Willem Buiter, 1990). However, this conventional view has been increasingly challenged in the context of developing economies, where public investment can play a catalytic role in stimulating private sector activity. For instance, António Afonso and Miguel St. Aubyn (2010) argue that public investment enhances productivity and growth expectations, thereby crowding in private investment. Similarly, Erkan Atukeren (2010) highlights that in emerging economies with infrastructure deficits, public capital formation often complements private investment. Cross-country evidence by Hassan Ahmed and Stephen Miller (2000) further supports this view, showing that government expenditure on transport and communication tends to crowd in private investment in developing countries, while welfare-oriented expenditure may crowd it out.

Empirical evidence across countries, however, remains mixed and context-specific. In Turkey, Hüseyin Şen and Ahmet Kaya (2014) find that government expenditure crowds in private

investment, whereas fiscal deficits exert a crowding-out effect. In contrast, Bing Wang (2005) shows that in Canada, public expenditure on human capital complements private investment, while infrastructure spending may substitute for private sector activity. Evidence from Brazil and China also reveals an asymmetric relationship, where public investment crowds out private investment in the short run but crowds it in over the long run (Bruno de Oliveira Cruz & José Teixeira, 1999; Dong Wu & Zhang Zhi, 2009). Meanwhile, studies on South Africa indicate an insignificant direct relationship between public and private investment, though indirect effects through macroeconomic channels persist (Uma Kollamparambil & Maria Nicolaou, 2011). In Pakistan, the composition of government expenditure is crucial, with infrastructure and social sector spending crowding in private investment, while defence and debt-related expenditure crowd it out (Aamir Hussain et al., 2009; Naeem Saeed et al., 2006).

The Indian evidence similarly reflects a nuanced and evolving relationship. Early studies suggest that public investment may crowd out private investment in the short run, primarily due to financial constraints, but this effect tends to dissipate in the long run as improved output expectations stimulate private sector activity (Pradeep Dash, 2016; Prakash Mitra, 2006). At a more disaggregated level, Luis Servén (1999) finds that public infrastructure investment crowds out private investment in the short run through credit rationing, but complements it in the long run by enhancing productivity. Using an asymmetric VAR framework, Lekha Chakraborty (2007) shows that the relationship between public and private investment depends critically on macroeconomic conditions such as exchange rates and fiscal deficits. More recent studies emphasize the role of economic reforms and structural changes, with Gaurav Bahal, Mehdi Raissi, and Volodymyr Tulin (2018) finding evidence of crowding-in effects in the post-reform period. Further, employing a non-linear ARDL approach, Nazia Akber, Manish Gupta, and Kishor Ranjan Paltasingh (2020) highlight the presence of asymmetric dynamics, with stronger complementarity between public and private investment under favourable macroeconomic conditions.

A related strand of literature focuses on the role of fiscal deficits in shaping private investment. Theoretically, higher fiscal deficits may crowd out private investment by increasing borrowing costs or reducing the availability of credit to the private sector. Empirical studies largely corroborate this view. Mahmoud Mahmoudzadeh et al. (2017) find that fiscal deficits crowd out private investment modestly in developed economies, while Anyanwu Augustine et al. (2018) report stronger effects in developing and oil-dependent economies. Country-specific studies further confirm significant crowding-out effects of fiscal deficits in South Africa (Rumbidzai Biza et al., 2015), India (Lekha Chakraborty, 2016; Pradeep Dash, 2016), China (Dong Wu & Zhang Zhi, 2009), and Pakistan (Aamir Hussain et al., 2009). Additionally, macroeconomic variables such as trade openness and output cyclicality have been found to positively influence private investment in developing economies (Davide Furceri & Ricardo Sousa, 2011).

Overall, the empirical literature does not yield a uniform conclusion regarding the crowding-out hypothesis. The relationship between public and private investment is contingent upon factors such as the composition of government expenditure, macroeconomic stability, financial conditions, and the stage of economic development. Moreover, differences in empirical methodologies—ranging from VAR and panel models to ARDL and non-linear frameworks—contribute to the diversity of findings.

Within this context, the study by Shankar & Trivedi (2021) – Government fiscal spending and crowd-out of private investment: Evidence for India offers a comprehensive and methodologically robust examination of the Indian case. Using an ARDL cointegration framework and disaggregated data across aggregate, sectoral, and industry levels, the study finds strong evidence of complementarity between public and private investment, while also highlighting the adverse effects of fiscal deficit, inflation expectations, and macroeconomic vulnerability on private investment.

Building upon this foundation, the present study extends the analysis by employing a similar empirical framework and revisiting the crowding-in versus crowding-out dynamics in India using updated data up to 2023–24. By doing so, it seeks to reassess whether the complementary relationship identified in earlier work continues to hold in the evolving macroeconomic environment, thereby offering a contemporary perspective on the role of fiscal policy in shaping private investment behaviour.

4. Empirical Methodology

This study employs two complementary econometric frameworks to examine the long-run and short-run dynamics of private corporate investment in India. The primary methodology is the Autoregressive Distributed Lag (ARDL) bounds-testing approach to cointegration developed by Pesaran, Shin, and Smith (2001). The secondary framework, used as a robustness check, is a Dynamic OLS / Autoregressive Distributed Lag (ADL) model estimated via Ordinary Least Squares (OLS). Both approaches are estimated over the period 1980–2023 (44 annual observations) and share the same set of variables, structural dummies, and reform-interaction terms.

4.1 Data and Variable Construction

All nominal investment and financial variables are deflated by the GDP deflator (base year 2011–12 = 100) to obtain real values before taking natural logarithms. The lending rate (WALR), expected inflation (EINF), output gap (OG), trade openness (TO), and sovereign macroeconomic vulnerability (SMV) enter in levels as they are I(0) or weakly trending. The stationarity properties of all variables were verified using the Augmented Dickey–Fuller (ADF), Phillips–Perron (PP), and Zivot–Andrews (1992) endogenous structural break unit root tests. All variables are confirmed to be either I(0) or I(1); no variable is I(2), satisfying the precondition for ARDL bounds-testing.

The gross fiscal deficit (GFD) is sourced from MoSPI and RBI DBIE. The public and private sector GFCF data, including industry-level disaggregations, were graciously provided by Shiv Shankar (RBI) and correspond to the back-series national accounts data with 2011–12 as the base year, as described in Shankar and Trivedi (2021). The net non-food credit flow (NCF) is sourced from the RBI Database on Indian Economy (DBIE). Expected inflation (EINF) is estimated as the HP-filtered trend component of CPI, augmenting IMF inflation forecasts to mitigate end-sample bias. The output gap (OG) is the principal component of four filter-based OG estimates (HP, BP-BK, and two BP-CF variants), following the approach of Shankar and Trivedi (2021). Sovereign macroeconomic vulnerability (SMV) is computed as the three-year rolling standard deviation of the cyclical component of a comprehensive vulnerability indicator combining fiscal deficit, current account deficit, and GDP deflator, following Becker and Mauro (2006).

4.2 The ARDL Bounds-Testing Framework (Primary Methodology)

The ARDL framework is particularly suited to investment analysis in India for three reasons. First, it accommodates regressors integrated of different orders, I(0) or I(1), without requiring pre-classification of variables as purely stationary or unit-root processes. Second, it estimates short-run and long-run dynamics simultaneously within a single-equation unrestricted error-correction model (ECM), providing the ECT coefficient as a formal test of long-run causality and adjustment speed. Third, it is more efficient than classical cointegration methods (Johansen, Engle–Granger) in small annual samples. The comprehensive ARDL unrestricted ECM specification is:

Equation (1) — General ARDL-ECM Specification:

$$\Delta \ln PRC_t = \mu + \Sigma \beta_{1i} \Delta \ln PRC_{t-i} + \Sigma \beta_{2i} \Delta \ln PS_{t-i} + \Sigma \beta_{3i} \Delta \ln GFD_{t-i}$$

$$+ \Sigma \beta_{4i} \Delta \ln NCF_{t-i} + \Sigma \beta_{5i} \Delta WALR_{t-i} + \Sigma \beta_{6i} \Delta EINF_{t-i}$$

$$+ \Sigma \beta_{7i} \Delta OG_{t-i} + \Sigma \beta_{8i} \Delta TO_{t-i} + \Sigma \beta_{9i} \Delta SMV_{t-i}$$

$$+ \varphi_1 \ln PRC_{t-1} + \varphi_2 \ln PS_{t-1} + \varphi_3 \ln GFD_{t-1} + \varphi_4 \ln NCF_{t-1}$$

$$+ \varphi_5 WALR_{t-1} + \varphi_6 EINF_{t-1} + \varphi_7 OG_{t-1} + \varphi_8 TO_{t-1} + \varphi_9 SMV_{t-1}$$

$$+ \rho \, Dummy_i + \varepsilon_t \quad ...(1)$$

where Δ is the first-difference operator; β coefficients capture short-run dynamics; level terms (φ) test for a long-run cointegrating relationship; and ε_t is a white-noise error term. The null hypothesis of no cointegration is H_0: $\varphi_1 = \varphi_2 = ... = \varphi_9 = 0$, tested using the Wald (bounds) F-statistic against the Pesaran et al. (2001) asymptotic critical value bounds. If the computed F-statistic exceeds the upper critical bound I(1), cointegration is confirmed; if it falls below the lower bound I(0), it is rejected; if it falls within the band, the result is inconclusive. Following confirmation of cointegration, the short-run ECM is estimated as:

Equation (2) — Short-Run Error Correction Model:

$$\Delta \ln PRC_t = \alpha + \Sigma \beta_i \Delta [Regressors]_{t-i} + \lambda ECM_{t-1} + \rho \, Dummy_i + \vartheta_t \quad ...(2)$$

where ECM_{t-1} is the one-period lagged residual from the long-run ARDL relationship and $\lambda \in (-1, 0)$ is the error-correction coefficient. A significant negative λ confirms long-run causality running from the regressors to private investment and measures the annual speed of adjustment back to equilibrium following a short-run shock.

4.3 ARDL Bounds-Testing Results

4.3.1 Aggregate-Level ARDL Results

Four aggregate ARDL models (M1–M4) were estimated with ln PRC as the dependent variable. Common regressors across all models are ln PS, ln GFD, ln NCF, WALR, and EINF. The models differ in their additional controls and the inclusion of the reform interaction term.

Model	Lag Order	Additional Controls	F-stat	Cointegration
M1	2,0,0,1,1,2,0,0	OG, TO	2.4445	Not confirmed

M2	2,1,0,1,1,2,0,0	TO, SMV	2.5697	Not confirmed
M3	2,2,0,1,1,2,0,0	OG, SMV	2.0371	Not confirmed
M4	2,0,0,1,2,2,0,0	SMV, DREF×lnPS	3.6396 **	Confirmed

Models M1–M3 yield F-statistics of 2.44, 2.57, and 2.04, all falling below the I(0) lower bound at the 5 per cent level, so the null of no cointegration cannot be rejected. Model M4, which includes the sovereign macroeconomic vulnerability variable and the post-FRBM reform interaction term (DREF × ln PS), produces an F-statistic of 3.64 ($p = 0.037$), which exceeds the I(1) upper bound and confirms a long-run cointegrating relationship among the variables.

This result is itself a substantive finding rather than a limitation. The cointegration being confirmed exclusively in the specification that models the post-reform regime interaction indicates that the long-run equilibrium relationship between public and private investment in India is regime-contingent: it becomes stable and recoverable as a cointegrating vector only when the structural shift in fiscal governance following the FRBM Act (2003) is explicitly accommodated. This is consistent with the broader narrative that the complementarity between public and private investment in India is a post-reform phenomenon, as documented by Bahal et al. (2018).

Variable	M1	M2	M3	M4 (Cointegrated)	A Priori Sign
ln PS (Public Investment)	−0.120	−0.896	+0.171	+0.059	+
ln GFD (Fiscal Deficit)	−0.635	−0.689	−0.487	−0.381†	−
ln NCF (Credit Flow)	+1.104†	+1.529†	+0.762	+0.724*	+
WALR (Lending Rate)	+0.075	−0.051	+0.155	+0.112	−
EINF (Expected Inflation)	−0.056	−0.155	+0.051	+0.153	−
OG (Output Gap)	+0.004	—	+0.013	—	±
TO (Trade Openness)	−0.026	−0.044	—	—	+/−
SMV (Macro Vulnerability)	—	+0.062	+0.002	+0.008	−
DREF × ln PS (Reform Interaction)	—	—	—	+0.050**	+

The fiscal deficit coefficient is negative and consistently signed across all models, providing robust support for the crowding-out hypothesis. In M4 (the only cointegration-confirmed specification), the long-run coefficient of −0.381 ($p \approx 0.096$) implies that a one per cent sustained increase in the gross fiscal deficit reduces private corporate investment by

approximately 0.38 per cent in the long run. This is broadly consistent with Shankar and Trivedi (2021), whose comparable coefficient exceeds 50 basis points, with the modest attenuation plausibly attributable to the extended sample incorporating the post-2019 period of fiscal expansion under the National Infrastructure Pipeline. The net credit flow coefficient (+0.724, p = 0.029) confirms credit availability as a key long-run enabler of private capital formation. Most significantly, the reform interaction term DREF × ln PS (+0.050, p = 0.023) represents a central finding of this study: in the post-FRBM era, a one per cent increase in public investment generates an additional 0.05 per cent in private investment, over and above the unconditional baseline. This constitutes regime-contingent crowding-in — a finding that underscores the importance of fiscal governance frameworks in shaping the investment complementarity.

Variable	M1	M2	M3	M4
Δ ln PRC (t−1)	0.187	0.201	0.252†	0.136
Δ ln PS	—	0.181	0.164	—
Δ ln NCF	0.681*	0.884*	0.409	1.188***
Δ WALR	−0.013	−0.022	−0.010	−0.009
Δ EINF	−1.451	−2.287*	+0.026	+0.677
Δ EINF (t−1)	+2.039†	+2.190*	+0.928	+0.070
ECT (λ)	−0.417***	−0.430***	−0.485***	−0.626***
Adj. R^2	0.358	0.370	0.299	0.460

The ECT coefficient in M4 is −0.626 ($p < 0.001$), implying that approximately 62.6 per cent of any short-run deviation from the long-run equilibrium in private investment is corrected within one year. This is a moderate but robust speed of adjustment, consistent with the partial adjustment nature of investment decisions and the typical inertia in capital expenditure planning in India. The ECT coefficients in M1–M3 (−0.417 to −0.485) are also significant, though these models do not satisfy the cointegration precondition. The contemporaneous credit flow term (Δ ln NCF = 1.188, $p < 0.001$ in M4) is the single strongest short-run determinant of private investment, confirming that credit availability is a binding short-run constraint on private capital formation in India.

4.3.2 Sectoral-Level ARDL Results

Three sectoral ARDL models replace the aggregate public investment variable with sector-specific measures: infrastructure (S1: ln INFRA), non-infrastructure (S3: ln NINFRA), and core infrastructure (S5: ln CNFRA), with a uniform lag structure of (2,0,0,1,2,2,0). The bounds F-statistics for the sectoral models range from 3.04 to 3.28, falling within or at the boundary of the inconclusive band (I(0) lower bound = 2.27; I(1) upper bound = 3.28 for k = 6 at 5%). The sectoral cointegration results are therefore weaker than the aggregate M4 finding, reflecting

the greater volatility and structural divergence in sector-specific investment series over the extended sample.

Irrespective of the sectoral classification of public investment, the fiscal deficit crowding-out coefficient is negative and statistically significant across all sectoral models, ranging from −0.582 (non-infrastructure) to −0.696 (infrastructure). This is the most robust sectoral finding. Credit flow is strongly positive across all sectoral models. The sectoral public investment coefficients themselves are statistically insignificant at conventional levels, suggesting that, at the sectoral level, the composition of public investment does not generate a significant independent crowding-in effect once fiscal balance and credit conditions are controlled for over the full 1980–2023 sample. This result contrasts with Shankar and Trivedi (2021), who find significant sectoral crowding-in, and reflects the structural divergence between the public and private investment trajectories in the 2019–2023 sub-period.

4.3.3 Industry-Level ARDL Results

Seven industry-level ARDL models were estimated with a fixed lag structure of (1,1,1,0,1,1,2). None of the seven models achieves confirmed cointegration at the 5 per cent level; F-statistics range from 0.94 to 1.81, substantially below the I(1) bound of 3.28 for k = 6. The failure to confirm cointegration at the industry level is attributed to the considerably greater volatility of industry-specific investment data relative to aggregate data, the multiple structural breaks spanning the 44-year sample, and the post-2019 divergence between public infrastructure expansion and subdued private investment. Long-run coefficient estimates at the industry level are accordingly treated as indicative only. Despite the absence of confirmed cointegration, the fiscal deficit crowding-out coefficient is negative and significant in most industry models (ranging from −0.607 in Construction to −0.826 in Trade, Hotels, and Restaurants), and credit flow is uniformly positive.

5. Robustness Check: Dynamic OLS / ADL(1) Framework

To validate the ARDL findings and address the cointegration challenges arising from the extended sample and multiple structural breaks, this study employs the Dynamic OLS / Autoregressive Distributed Lag (ADL) model as a systematic robustness check. The ADL(1) framework avoids the requirement of a formal cointegration pre-test by recovering long-run effects directly via the Koyck transformation. It is particularly robust in samples characterised by multiple structural breaks and industry-level noise, as the long-run multiplier is recovered as a simple ratio of short-run coefficients and does not depend on the existence of a formal cointegrating vector.

5.1 ADL(1) Model Specification

The general ADL(1) aggregate specification is:

Equation (3) — ADL(1) Aggregate Specification:

$$\ln PRC_t = \alpha_0 + \rho \cdot \ln PRC_{t-1} + \beta_1 \cdot \ln PS_t + \beta_2 \cdot \ln GFD_t + \beta_3 \cdot \ln NCF_t$$

$$+ \beta_4 \cdot WALR_t + \beta_5 \cdot EINF_t + \beta_6 \cdot SMV_t$$

$$+ \delta_1 \cdot DBREAK + \delta_2 \cdot (DREF \times \ln PS_t) + \varepsilon_t \quad ...(3)$$

where $\ln PRC_{t-1}$ captures partial adjustment dynamics with $\rho \in (0, 1)$. The long-run (steady-state) multiplier for any regressor X is recovered as:

Equation (4) — Long-Run Multiplier:

Long-Run Effect of X = $\hat{\beta}_i / (1 - \hat{\rho})$ ...(4)

The dummy variable DBREAK equals 1 from 2003 onward, endogenously identified by the Zivot–Andrews test as the structural break in ln PRC coinciding with the FRBM Act. The interaction term DREF × ln PS tests whether the public–private investment relationship strengthened in the post-reform regime. For sectoral and industry-level models, ln PS is replaced by the respective sectoral or industry public investment variable while the interaction term retains the aggregate reform definition.

5.2 Aggregate ADL(1) Results

The fitted aggregate ADL(1) equation is:

$$\ln PRC_t = 11.030 + 0.418 \cdot \ln PRC_{t-1} - 0.134 \cdot \ln PS_t - 0.290 \cdot \ln GFD_t$$

$$+ 0.147 \cdot \ln NCF_t - 0.009 \cdot WALR_t + 0.037 \cdot EINF_t + 0.007 \cdot SMV_t$$

$$+ 0.250 \cdot DBREAK + 0.021 \cdot (DREF \times \ln PS_t) + \varepsilon_t$$

$R^2 = 0.914$; Adj. $R^2 = 0.890$; $F(9,33) = 38.84$ ($p<0.001$); RSE = 0.1199

Variable	Coeff.	Std. Err.	t-stat	p-value	Sig.	LR Effect
Intercept (α_0)	11.030	3.773	2.923	0.006	**	—
ln PRC (t−1) [ρ]	0.418	0.129	3.250	0.003	**	—
ln PS [β_1]	−0.134	0.242	−0.554	0.583	ns	−0.230
ln GFD [β_2]	−0.290	0.112	−2.596	0.014	*	−0.498
ln NCF [β_3]	+0.147	0.100	1.461	0.153	ns	+0.252
WALR [β_4]	−0.009	0.021	−0.420	0.677	ns	−0.015
EINF [β_5]	+0.037	0.042	0.870	0.391	ns	+0.063
SMV [β_6]	+0.007	0.026	0.261	0.796	ns	+0.012
DBREAK [δ_1]	+0.250	0.129	1.938	0.061	†	+0.429
DREF × ln PS [δ_2]	+0.021	0.009	2.340	0.025	*	+0.036

The partial adjustment coefficient $\rho = 0.418$ ($p = 0.003$) confirms investment inertia: approximately 58.2 per cent of the gap between actual and desired investment is closed within a year, consistent with the ECT-implied 62.6 per cent from the ARDL model. The fiscal deficit

exerts a significant negative short-run effect (−0.290, p = 0.014), translating to a long-run crowding-out of −0.498 — closely aligned with the 50 basis point estimate of Shankar and Trivedi (2021). The structural break dummy DBREAK (+0.250, p = 0.061) confirms a positive level shift in private investment from 2003 onward, corresponding to the improved fiscal governance under the FRBM Act. The reform interaction term (DREF × ln PS = +0.021, p = 0.025; LR: +0.036) corroborates the ARDL finding, confirming that economic reforms strengthened the complementarity between public and private investment in the post-liberalisation period. All diagnostic tests pass: the Breusch–Godfrey test (p = 0.348) confirms no serial correlation, the Ramsey RESET test (p = 0.120) confirms correct functional form, and CUSUM plots confirm parametric stability throughout the sample.

5.3 Sectoral and Industry-Level ADL(1) Results

The ADL robustness check yields consistent results across all six specifications. The fiscal deficit crowding-out coefficient ranges from −0.487 to −0.490 in long-run terms at the sectoral level, and from −0.457 to −0.546 at the industry level. These estimates are stable and precisely estimated, providing strong confirmation that the fiscal crowding-out finding is not an artefact of the ARDL specification or the cointegration requirement. The reform interaction term (DREF × ln PS) is positive and significant across all ADL specifications, ranging from +0.027 to +0.041 in long-run terms, further confirming the regime-contingent crowding-in result. All six ADL models achieve adjusted R^2 values between 0.889 and 0.898 and pass all diagnostic tests, demonstrating excellent model fit and stability.

Finding	ARDL (M4)	ADL (Aggregate)	Convergence ?
Long-run fiscal crowding-out	−0.381 (10%)	−0.498 (5%)	Yes ✓
Post-reform interaction (DREF×lnPS)	+0.050 (5%)	+0.021 (5%)	Yes ✓
Structural break (DBREAK)	Confirmed (ZA)	+0.250 (10%)	Yes ✓
Credit flow (NCF) positive	+0.724 (5%)	+0.252 LR	Yes ✓
Speed of adjustment	ECT = −0.626 (63%/yr)	$\hat{\rho}$ = 0.418 (58%/yr)	Yes ✓
Public investment (ln PS) baseline	Insignificant	Insignificant (−0.230 LR)	Yes ✓

The convergence of both methodologies on all six key findings constitutes the strongest argument for the robustness of this study's conclusions. The near-identical fiscal deficit crowding-out coefficient (−0.38 vs. −0.50), the consistent sign and significance of the reform interaction term, and the comparable speeds of adjustment across the ARDL and ADL frameworks collectively confirm that the core findings are not methodology-dependent.

6. Summary of Findings and Policy Implications

This study's findings collectively support the following conclusions. First, fiscal deficit crowding-out of private corporate investment is a robust, long-run phenomenon in India, with a one per

cent increase in the gross fiscal deficit reducing private investment by approximately 38–50 basis points. This finding is consistent across all aggregate and sectoral specifications in both the ARDL and ADL frameworks, and across all seven industry-level specifications, confirming the robustness of the crowding-out hypothesis for India over the 1980–2023 period. The crowding-out magnitude is most pronounced in the Trade, Hotels, and Restaurants and Mining sectors, and somewhat attenuated for infrastructure-heavy sectors.

Second, unconditional public sector investment does not exert a significant independent crowding-in or crowding-out effect on private corporate investment at the aggregate level. This result contrasts with the findings of Shankar and Trivedi (2021) for the shorter 1981–2019 sample and reflects the structural divergence of the post-2019 period. However, conditional on the post-FRBM reform regime, public investment does exert a significant crowding-in effect: the reform interaction term is positive and significant in both the ARDL and ADL frameworks, with a long-run magnitude of +0.036 to +0.050.

Third, credit availability is the most powerful short-run determinant of private corporate investment. The contemporaneous bank credit flow coefficient in the ARDL ECM (1.188, $p < 0.001$) and the ADL long-run credit multiplier (+0.252) confirm that access to bank financing is a binding constraint on private capital formation in India. Policies that deepen financial intermediation and maintain adequate credit flows to the private corporate sector are therefore a necessary complement to fiscal consolidation.

Fourth, the 2003 structural break associated with the FRBM Act represents a significant positive regime shift for private investment, contributing approximately 25–43 per cent to the level of private investment in the post-2003 period. This finding underscores the long-term investment dividends of credible fiscal consolidation frameworks.

The policy implications are clear. Sustained fiscal consolidation is the most important lever for crowding-in private investment. Directing public expenditure toward infrastructure, which attenuates the adverse fiscal multiplier, combined with maintaining adequate credit flows to the private corporate sector and containing inflation and sovereign vulnerability, would create the most conducive environment for private capital formation. Quality fiscal adjustment — consolidation driven by expenditure rationalisation rather than investment compression — is preferable to deficit reduction achieved by cutting productive public capital spending.

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